International Journal of Information Systems in the Service Sector
Volume 14 • Issue 1

Topic Modeling and Sentiment Analysis of Yelp Restaurant Reviews

Sonya Zhang, California State Polytechnic University, Pomona, USA*
Linda Ly, California State Polytechnic University, Pomona, USA
Norman Mach, California State University, Pomona, USA
Christian Amaya, California State Polytechnic University, Pomona, USA

ABSTRACT

The use of online consumer reviews (OCRs) websites like Yelp or TripAdvisor has increasingly gained popularity. However, many products or services now have a large number of online reviews, which makes it difficult for consumers to decide which reviews to pay attention to or for businesses to identify which critical areas to improve on. The use of text mining and sentiment mining techniques is deemed important to automatically process the content of online reviews and help improve review usefulness. Applying a four-phase research model, the authors performed topic modeling using Latent Dirichlet allocation (LDA) to extract five topics (price, time, food, service, and location) and sentiment analysis using Python TextBlob to aggregate consumer sentiment per topic from a Yelp restaurant reviews dataset. The authors propose that the design of recommender systems for OCRs or business decision-making systems can be faster, simpler, and more useful by integrating automatic topics extraction and sentiment analysis.

KEYWORDS

Machine Learning, Sentiment Analysis, Text Mining, Topic Extraction, Topic Modeling, Yelp

INTRODUCTION

Crowd-sourced business review and social networking sites like Yelp and TripAdvisor keep gaining significant traffic, as they provide platforms for consumers to search information, discuss, and share purchase experiences, which generates electronic word-of-mouth (eWOM) including OCRs (Bilal et al., 2020). With close to 1 million daily visitors worldwide, Yelp.com is the 47th most visited website in the U.S. and No. 166 in the world in July 2021 according to SimilarWeb, a leading digital intelligence platform that provides web analytics services. This is because OCRs significantly impact consumers' purchase intention as well as businesses' improvement initiatives. Consumers today have easy access to OCRs to see how others had evaluated the products and services, which helps them to make informed purchase decisions (Nakayama & Wan, 2018, 2019). Insights from OCRs also help online business owners enhance the experience level of their customers (Dhahak & Huseynov, 2020). Businesses with free Yelp accounts can see an average of $8,000 in annual revenue and $23,000 in

DOI: 10.4018/IJISSS.295872 *Corresponding Author

Copyright © 2022, IGI Global. Copying or distributing in print or electronic forms without written permission of IGI Global is prohibited.

annual revenue through advertisements (Bassig, 2018). However, the large volume of reviews on numerous products and services can be overwhelming and resource intensive for businesses and customers to extract useful information and make timely decisions.

There are several directions of research that aim to solve this problem. Prior research studied the factors that influence restaurant reviews, sentiment, and ratings (Baek, Ahn and Choi 2012; Cao, Duan and Gan 2011; Salehi-Esfahani & Kang, 2019.) Knowing and understanding these influential factors can help summarize and pinpoint valuable and critical insights (Pezenka & Weismayer, 2020; Tian, Lu, & McIntosh, 2021.) Another direction of research is using text mining (e.g., topic modeling, classification, and clustering) and sentiment mining techniques to examine how to use the text content of the review to predict its usefulness and helpfulness automatically and accurately (Aghakhani, Salehan, & Zhang, 2017; Ajrawi et al., 2021; Li et al., 2019; Salehan, Zhang, & Aghakhani, 2017) Lastly, prior studies used supervised or unsupervised learning to detect fake reviews and understand what makes people trust certain reviews (Elmogy et al., 2021.)

However, most existing research studies predicting the review helpfulness from review text were limited to a single machine learning model and the scope of their sample dataset. This made their findings lack generalization or practical implications. The novel approach and findings in this study is a four-phase research model demonstrating how to develop topic modeling and sentiment analysis models that help businesses and consumers discover and aggregate insights from a large number of online consumer reviews in a faster, simpler, and more reliable way.

LITERATURE REVIEW

In the following, we analyzed and evaluated relevant works in online consumer reviews (OCR), review helpfulness, Natural Language Processing (NLP), topic modeling, and sentiment analysis.

Online Consumer Reviews (OCR)

Consumer reviews are considered a powerful business intelligence and performance measurement for businesses. Online reviews have the power to gain customer trust (Pranata & Susilo, 2016; Zhu, Lin, & Cheng, 2020) and can strengthen a company's credibility (Chakraborty & Bhat, 2018). According to a survey conducted by TripAdvisor, 94% of U.S. diners are influenced by online reviews. Another survey by Small Biz Trends (June 2018) found that 31% of customers said they would likely spend more money on businesses with positive reviews, whereas 22% said they would not spend money at a business after reading one negative review. Customer reviews also help businesses identify critical areas for improvement (Zhang et al., 2018).

Online review websites and apps such as Yelp or TripAdvisor have made it easier for consumers to post reviews about a business and to access reviews posted by others. These online review websites and apps are essentially vast online warehouses of unstructured data, as millions of reviews are already published by consumers voluntarily and continue to grow every second. However, the large number of reviews can make it difficult for consumers and businesses to decide which reviews are more significant and the overall trend or pattern (Zhang et al., 2018). Currently, Yelp's user interface lacks a sophisticated filtering function for search results. At the time of this study, the available filtering function includes basic keyword search and sort, as seen in Figure 1.

Big data, machine learning, and Natural Language Processing (NLP) have enabled a whole new level of knowledge discovery about products and services for both businesses (Dumont, 2017; Ting et al., 2017) and consumers (Zhang et al., 2018).

Table 1 shows the review attribute, methodology, citation, and main findings from prior studies. The common review attributes most important to consumers are (ordered by frequency from the most to the least): Product, Service, Price, Location, Atmosphere (or Ambience), Cleanliness, Menu, and Hours. Topic modeling or topic extraction from online consumer reviews can benefit both consumers making purchasing decisions and businesses looking for areas it can improve on.

14 more pages are available in the full version of this document, which may be purchased using the "Add to Cart" button on the product's webpage:
www.igi-global.com/article/topic-modeling-and-sentiment-analysis-of-yelp-restaurant-reviews/295872?camid=4v1

Related Content

Stakeholder Ontology and Mining for Improving Complex Services
Jay Ramanathan and Rajiv Ramnath (2013). *International Journal of Information Systems in the Service Sector (pp. 65-79).*
www.igi-global.com/article/stakeholder-ontology-mining-improving-complex/78937?camid=4v1a

Engineering Financial Enterprise Content Management Services : Integration and Control
Dickson K. W. Chiu, Patrick C. K. Hung and Kevin Kwok (2010). *International Journal of Systems and Service-Oriented Engineering (pp. 86-113).*
www.igi-global.com/article/engineering-financial-enterprise-content-management/44688?camid=4v1a

Adoption of a Comprehensive Web-Based Wealth Management Service
Tomi Dahlberg and Anssi Öörni (2010). *International Journal of E-Services and Mobile Applications (pp. 1-18).*
www.igi-global.com/article/adoption-comprehensive-web-based-wealth/46069?camid=4v1a

The Economics of Cloud Computing
Federico Etro (2013). *Cloud Computing Service and Deployment Models: Layers and Management (pp. 296-309).*
www.igi-global.com/chapter/economics-cloud-computing/70148?camid=4v1a